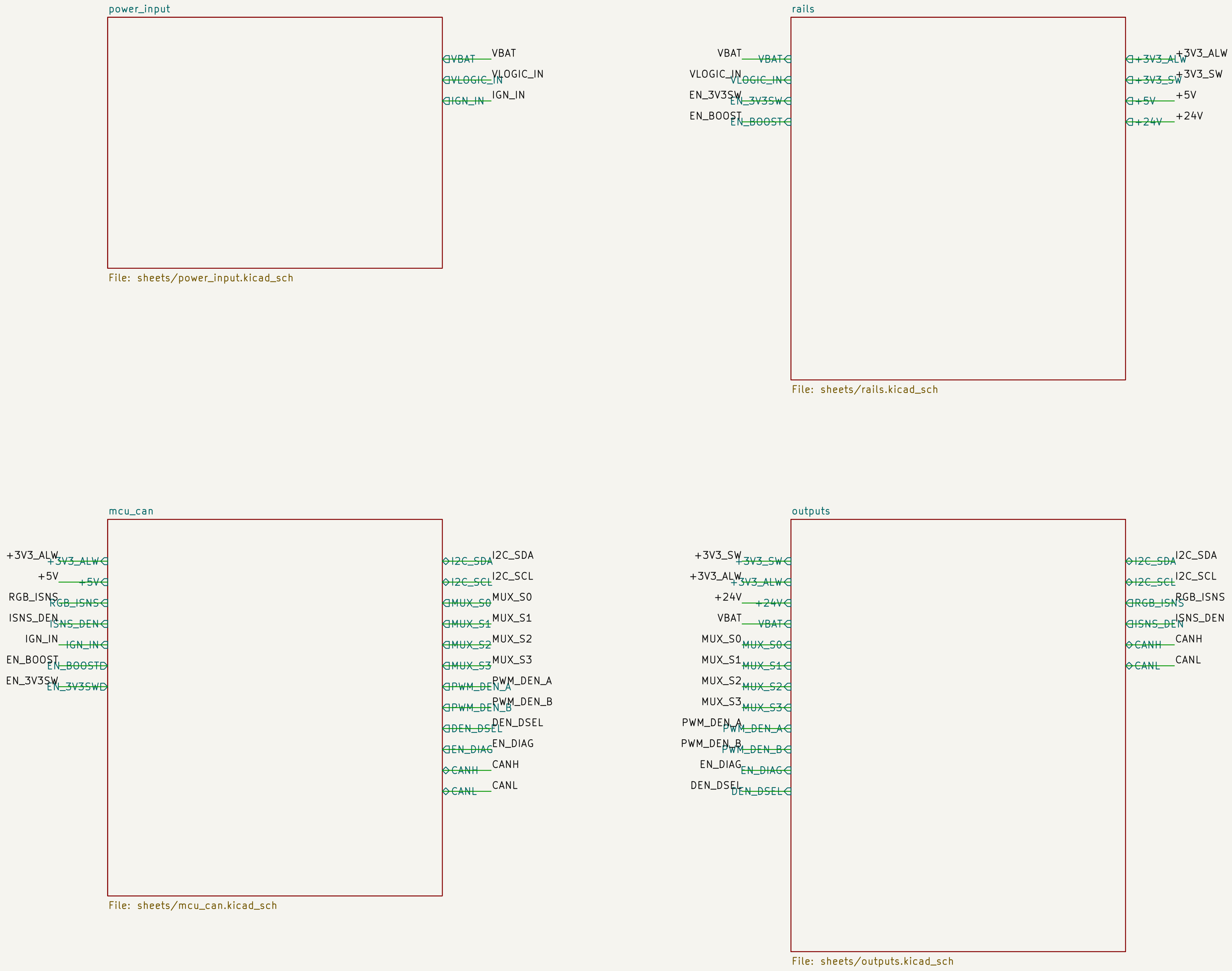
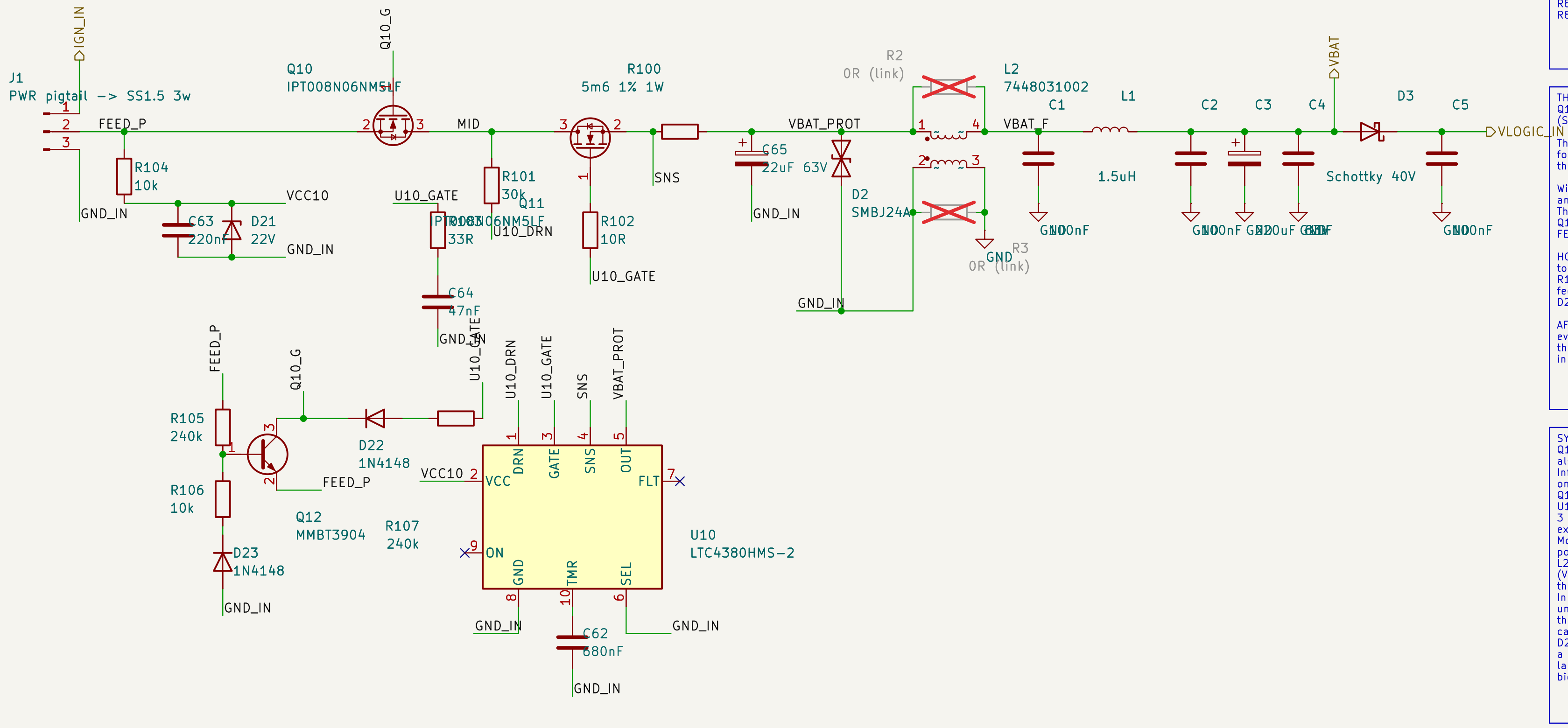


Sheet pins are joined by root-level labels, not by wires between boxes
Net naming contract: see hardware/README.md

Sheet: /
File: mccan.kicad_sch

Title: MCCAN Root

Size: A3 Date:
Rev:
KiCad E.D.A. 10.0.2
Id: 1/5



COMMON-MODE CHOKE AND THE SPLIT GROUND -- DO NOT "TIDY" THIS

GND_IN (connector side) and GND (board side) are SEPARATE NETS, joined only through choke winding B. Nothing else may bridge them -- no wire, no capacitor, and on the PCB no ground-plane copper.

The usual way this gets broken is CAPACITIVE, not a wire: a decoupling cap from VBAT_PROT (upstream) to GND (downstream) bridges winding A's input to winding B's output, giving HF a path straight around the choke. The circuit then works perfectly and filters nothing, and no ERC or DRC check will complain.

That is why C1 sits on VBAT_F (downstream) and the TVS anode sits on GND_IN (upstream).

THE ENTIRE eFUSE SITS ON GND_IN. U10's GND and SEL, C62, C63, C64, C65 and D21/D23's anodes all return to the connector side, because U10 measures the feed and must reference the return the battery is actually connected to. Do not move any of them to board GND to "shorten the return path".

BUILD OPTION: fit EITHER L2 (the choke) OR R2+R3 (0R links that short the windings). Links give the simple tied-ground case for the first prototype; the footprint is present either way so fitting a real choke later needs no board revision.

LAYOUT (Task 8): the ground plane must not bridge the choke.

GND_IN is a LOCAL label, never a power symbol. A second power symbol named GND would merge the two returns globally and silently defeat the whole arrangement.

THE eFUSE (U10) -- WHAT REPLACED F1, Q1, D1 AND R1

This front end used to be a 10 A blade fuse, a P-channel reverse-polarity FET, a gate Zener and a 1M gate pull-down. Spec 4.2 replaced all four with ONE element: U10, an LTC4380-2 surge stopper driving a back-to-back pair Q10/Q11. It does three jobs the old arrangement did not do together --

- REVERSE POLARITY, by the opposed body diodes of Q10/Q11 (see the common-drain note below).
- OVERCURRENT, as an active current limit across R100 (5.6 mOhm), with a C62-timed fault trip and the -2 suffix's AUTO-RETRY at roughly 0.1-0.2% duty -- about one retry every 1.5-3 s. Auto-retry is required, not preferred: the MCU is powered downstream of this stage and could never clear a latch.
- OVERVOLTAGE, by holding VBAT_PROT near 29.5 V through a linear clamp. That is what the internal 31.5 V gate clamp gives with SEL tied to GND_IN.

The reference circuit is the datasheet's FIGURE 7 (12 V overcurrent-protected high-side switch), NOT Figure 5. Figure 7 is 12 V in, back-to-back FETs, an LTC4380-2, 80 VDC max against our 50.6 V worst case. Only R100 (R_SNS), R101 (R_DRN) and C62 (C_TMR) are rescaled -- Figure 7 is a 2 A design.

TWO DELIBERATE DEPARTURES FROM FIGURE 7, both checked:

- R103/C64 are 33 Ohm / 47 nF, where Figure 7 draws 150 Ohm / 0.1 uF. 33 Ohm / 47 nF is the datasheet's own stated MINIMUM in PIN FUNCTIONS, and it is chosen because a larger C64 lengthens the output ramp: 0.1 uF pushes the fault timeout past 10 ms, off the end of the SOA curve that was checked. Do not fit a larger C64 without re-checking the FETs' safe operating area.
- ON is left open on its internal 1 MOhm pull-up and FLT is not routed to the MCU. Both carry no-connect flags so ERC stays clean; both are available if a later revision wants them.

C65 IS NOT OPTIONAL. PIN FUNCTIONS requires a minimum of 22 uF bypassing OUT "as close to the pin as possible". Nothing else on this sheet satisfies it -- C1 is 100 nF and sits behind the choke, C3 sits behind L1 as well. Layout must keep C65 next to U10 pin 5, not merely on the same net.

REFERENCE DESIGNATORS: spec 4.2 names these parts R88-R95, C49-C52 and D22-D24. Those are taken -- R88-R99 and C49-C61 are the RGB output snubbers on the outputs sheet. The mapping actually built is R88->R100 .. R95->R107, C49->C62 .. C52->C65, and D22->D21 / D23->D22 / D24->D23.

THE PASS PAIR IS COMMON DRAIN. DO NOT "SIMPLIFY" IT.

Q10 pin 3 (DRAIN) and Q11 pin 3 (DRAIN) meet at MID. Q10 pin 2 (SOURCE) goes to FEED_P; Q11 pin 2 (SOURCE) goes to R100 and on to VBAT_PROT. Sources OUTWARD, drains tied in the middle.

That is what makes the body diodes OPPOSE each other, and it is the entire reverse-polarity block -- forward current reaches the load through Q11's channel, reverse current is stopped by Q10's diode facing the wrong way, and that holds with the part unpowered.

Wired common SOURCE instead, the board works normally, passes ERC, passes DRC, passes the pad count, and has no reverse-polarity protection at all. An earlier draft of spec 4.2 instructed exactly that. The tell that it is wrong is the gate network: under common source both gates would sit on one node and Q12/D22/D23/R105-R107 would have nothing to do. They exist because Q10's gate must be referenced to FEED_P while U10's GATE pin is referenced to OUT.

HOW Q10'S GATE IS DRIVEN. U10's GATE sits at about OUT + 11.5 V. R107 (240k) and D22 carry that across to Q10's gate, landing it near FEED_P + 10.8 V -- enough to enhance a FET whose source is at the input. R105 (240k from FEED_P) biases Q12's base through R106/D23; under reverse polarity GND_IN rises above the feed, D23 conducts, Q12 turns on and clamps Q10's gate to its own source, holding Q10 hard off. D23 stands off up to 50.6 V in normal operation, which is why it is a 75 V part and not a Schottky.

AFTER ANY CHANGE HERE, READ THE NETLIST BY HAND. gen_power_input.py prints the expected coordinate for every pin of every net; spec 4.2 makes that check a formal acceptance criterion. This is the error class that has got through every automated check on this board three times now -- the CM choke's pin numbering in Task 7, the RGB FET clamp path in review finding F6, and the first draft of this very paragraph.

SYMBOL AND PIN-NUMBER NOTES (this sheet was authored as text, not drawn in the GUI)

Q10 and Q11 use Transistor_FET:Q_NMOS_GSD, which numbers the pins 1 = G, 2 = S, 3 = D. Device:Q_NMOS also exists but numbers its pins with the LETTERS D / G / S, which map to no stock footprint. The Infineon HSOF-8 footprint numbers only three pads, 1/2/3, which is what lets the three-pin symbol map onto an eight-lead package.
Q12 uses Transistor_BJT:Q_NPN_BEC -- 1 = B, 2 = E, 3 = C -- matching a SOT-23 MMBT3904 pinout.
U10 uses mccan_parts:LTC4380, built by symlib.py from the datasheet's PIN CONFIGURATION: 1 DRN, 2 VCC, 3 GATE, 4 SNS, 5 OUT, 6 SEL, 7 FLT, 8 GND, 9 ON, 10 TMR. That is the MS (MSOP-10) package, which has NO exposed pad. The DD package does, numbered 11; ordering DD would leave the pad unlanded.
Most of U10's pins are typed PASSIVE on purpose. The part sits on GND_IN, which by design carries no power symbol, so a power_in GND or VCC pin would make ERC demand a driver that must never exist.
L2 uses mccan_parts:CM_CHOKE_4T, not Device:L_Coupled. Winding A is pins 1-4 and carries the FEED (VBAT_PROT -> VBAT_F); winding B is pins 2-3 and carries the RETURN (GND_IN -> GND). L_Coupled numbers the same part 1-2 / 3-4, and using it here shorted VBAT_PROT to GND through a winding.
In the KiCad 10 Device library a diode's pin 1 is the CATHODE (K) and pin 2 is the ANODE (A), and the unrotated symbol draws the cathode on the LEFT. D3, D21 and D23 are rotated here so their cathodes face the nodes named in the net table; check polarity by the K / A pin names, not by pin-number habits carried over from KiCad 9 or earlier.
D2 is Device:D_TVS, whose pins are A1 / A2 -- that is the BIDIRECTIONAL TVS symbol, while the SMBJ24A is a UNIdirectional part. Pin 1 is wired to VBAT_PROT so that pad 1 (the cathode band on an SMB body) lands on the rail. Either swap in a unidirectional symbol, or change the part number to SMBJ24CA if a bidirectional clamp is what is actually wanted.

VALUE NOTES

R100 = 5.6 mOhm, the current-limit sense. It carries the whole 7.0 A night load: 0.27 W, so 2512 at a minimum and a 4-terminal (Kelvin) part if one can be had. Task 8 must give it its own copper -- that 0.27 W couples into the board, not into the heat-spreader plate.
R101 = 30 kOhm sets the SOA multiplier from the voltage across the pair. It is taken to MID, not to FEED_P: it senses Q11's Vds. Sized from the DRN pin's 1 mA limit at the SMBJ24A's 50.6 V 8/20 us clamp -- (50.6 - 27) / 1 mA = 23.6 k, rounded up to the next standard value, giving 0.787 mA at 50.6 V.
C62 = 680 nF sets the fault timeout. Worst-case inrush ramp with C64 fitted is 95 ms at 2.4 uA, and the datasheet's guideline is to hold TMR's rise to 0.4 V during it. Fault timeouts land at 9.2 ms into a 12 V dead short and 8.6 ms in the 23.6 V clamp case, both inside Diagram 3 derated to a 65 C case.
R104 / C63 / D21 = 10k / 220 nF / 22 V is U10's VCC feed, hold-up and clamp. It appears in all three of the datasheet's application circuits including the 20 V one, so it is not a high-voltage artefact.
R102 = 10 Ohm damps Q11's gate. R103 / C64 = 33 Ohm / 47 nF compensate the current-limit amplifier -- POPULATED, not DNP; see the eFuse note for why the value is the datasheet minimum rather than Figure 7's.
R105 / R106 / R107 = 240k / 10k / 240k with Q12, D22 and D23 steer Q10's gate. Jellybeans, but R105 and R107 stand off the full 50.6 V clamp so they are 1206 rather than 0603.
R2 and R3 = 0R links. Fit EITHER L2 OR R2+R3, never both -- see the split-ground note.
D2 = SMBJ24A load-dump clamp. D3 = 40 V Schottky feeding the logic rail.
C3 = 220 uF electrolytic bulk; C1 / C2 / C4 / C5 ceramic; C65 = 22 uF, U10's mandatory output bypass.
L1 = 1.5 uH, the series element of the pi filter; L2 = the common-mode choke.

STILL OPEN: no orderable is named for Q12, D21, D22, D23 or C65, and the HSOF-8 variant for Q10/Q11 (-1 versus -2 versus -3) has not been confirmed against the Infineon package drawing.

INPUT FILTER ORDER: eFuse -> TVS -> CM choke -> pi filter -> bulk

The TVS (D2) must come BEFORE the choke. Surge clamping degrades with series inductance between the entry point and the clamp, so any inductance ahead of D2 lets the clamped voltage overshoot.
The choke (L2) must come BEFORE the pi filter. The common-mode noise being blocked originates INSIDE the board -- it is the boost converter on the rails sheet -- so the choke has to sit between the switcher and the harness, not out at the connector.
D2 sits DOWNSTREAM of the eFuse, on VBAT_PROT. That is deliberate and it is an assumption spec 4.2's SOA arithmetic depends on: the 23.6 V-at-7 A clamp case is computed with the TVS not clamping the input. Moving D2 to FEED_P would protect Q10/Q11 from the surge directly and is arguably better -- it is logged as an open question for spec 4.3, and it invalidates the SOA numbers if it is done without redoing them.

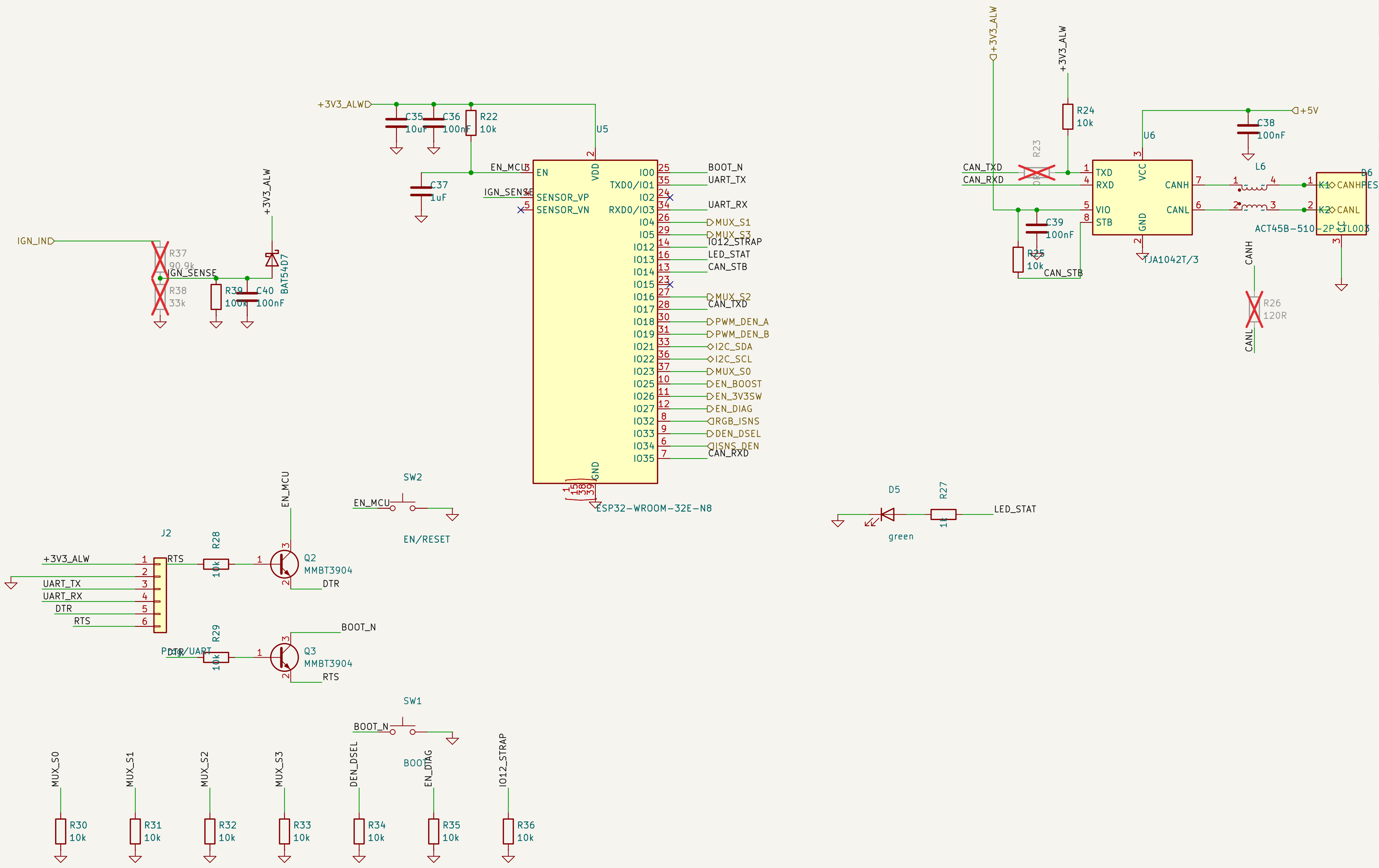
NOT DRAWN: THE DNP SECOND FEED AND SECOND SCHOTTKY (spec section 4.4)

Spec 4.4 asks for a second 12 V feed and a second Schottky as FOOTPRINTS ONLY, fitted Do-Not-Populate, so that an ideal-diode OR can be retrofitted without a board respin. Those parts (J2 / F2 on net VBAT_B, and a second Schottky from VBAT_B into VLOGIC_IN) are intentionally absent here. Add them as DNP symbols before PCB layout if you want them in the netlist -- a footprint that is not in the netlist will not be placed on the board.

Sheet: /power_input/
File: power_input.kicad_sch

Title: power_input

Size: A3 | Date: | Rev:
KiCad E.D.A. 10.0.2 | Id: 2/5



ACCEPTANCE (spec 7, 8.1)

- ESP32-WROOM-32E-N8 (8 MB). NOT the 4 MB variant, NOT the -32UE (U.FL) variant.
- Module powered from +3V3_ALW so it stays up in sleep.
- Pin assignment EXACTLY per the plan's Global Constraints table.
- CAN_RXD on GPIO 35 (RTC-capable, EXT0 wake); IGN_SENSE on GPIO 36 (EXT1 ANY_HIGH).
- ISNS_DEN on GPIO 34 (ADC1 ONLY); DEN_DSEL on GPIO 33 – the PROFET has ONE multiplexed IS output, not two. GPIO 39 is spare.
- CAN_STB on GPIO 14, PULLED UP to VIO (floating = standby).
- TJA1042T/3 TXD through an UNPOPULATED 0 ohm link, TXD pulled to VIO.
- NO 120 ohm terminator populated; unpopulated footprint present.
- CAN common-mode choke + ESD protection on CANH/CANL.
- Programming header with DTR/RTS auto-reset pair, BOOT and EN buttons.
- IGN_SENSE divider footprint present but UNPOPULATED.

HARDWARE-ENFORCED LISTEN-ONLY

R23 (0 ohm) is DNP. With it unpopulated the transceiver TXD is held recessive by R24 and the board is PHYSICALLY INCAPABLE of transmitting on the vehicle bus. Populate R23 only for deliberate bench TX.
This is not a firmware setting that can be got wrong, and not a fuse that can be blown by accident. The link is the only conductive path from GPIO 17 to the transceiver, so a board built to the BOM cannot assert a dominant bit on a moving motorcycle's bus no matter what the firmware does.

TWO SUPPLIES, ON PURPOSE – DO NOT TIE THEM TOGETHER

VIO (pin 5) -> +3V3_ALW, always on. VCC (pin 3) -> +5V, gated off in sleep.
The split is what makes CAN wake work. In sleep the +5V rail is down, so the transmit side and the bus drivers are unpowered, but VIO keeps the receiver and the RXD output alive. Bus traffic therefore still pulls RXD low, which is the EXT0 wake event on GPIO 35.
Powering both from one rail loses either the wake path or the sleep budget.
TJA1042T/3 VCC is 4.5–5.5 V. The "/3" suffix refers to VIO only. Running VCC from 3.3 V is out of spec – this was a genuine error in an early draft of the spec.

ANTENNA KEEP-OUT – carries into layout (Task 8)

No copper, no plating, no components and no ground pour may sit under the module's antenna area, on ANY layer. Overhang the module past the board edge, or cut the board back to it.
The finned/metal wall of the enclosure must not be the antenna wall either.

WHY THE AUTO-RESET PAIR IS CROSS-COUPLED

Each transistor's emitter sits on the OTHER control line, so the pair only acts when DTR and RTS differ. Both high or both low = run mode. That is what stops a plain serial monitor – which asserts one line on open – from dropping the board into the bootloader.
esptool drives DTR=0/RTS=1 to assert reset, then DTR=1/RTS=0 to hold IO0 low while EN releases. Q2 pulls EN low in the first state; Q3 pulls IO0 low in the second.

PASSIVE DEFAULTS (block F) – MANDATORY, NOT OPTIONAL

Every resistor in block F is required by the plan's biasing table. Two are boot requirements rather than merely safe defaults:
R33 on MUX_S3: GPIO 5 is a STRAPPING pin. It is sampled at reset and must be defined.
R36 on IO12_STRAP: GPIO 12 (MTDI) selects the flash voltage at boot. Sampled high it selects 1.8 V, which will not run the module's 3.3 V flash. Most WROOM-32E parts ship with the eFuse already burned so the strap is ignored, but that is not guaranteed across date codes and it is not worth a 10k resistor to find out. ADDED 2026-09-14 – this one is not in the plan; drop it if you would rather follow the plan exactly.
These draw ZERO current in sleep: the GPIO is driven low, so there is no potential across them.

IGNITION SENSE (block E) – divider DNP, pulldown POPULATED

R37 and R38 are DNP. R39 is NOT. GPIO 36 has no internal pull-up or pull-down and EXT1 ANY_HIGH wake is armed on it, so a floating pin wakes the board on noise and flattens the battery. R39 defines it low whether or not the divider is fitted.
If the divider is ever populated: 90.9k over (33k || 100k = 24.8k) gives 2.57 V at 12.0 V in and 2.96 V at 13.8 V, both above the 2.48 V input-high threshold, with D8 clamping anything above the rail. Current draw is 104 uA, but only while the ignition line is live, so it does not touch the parked budget.

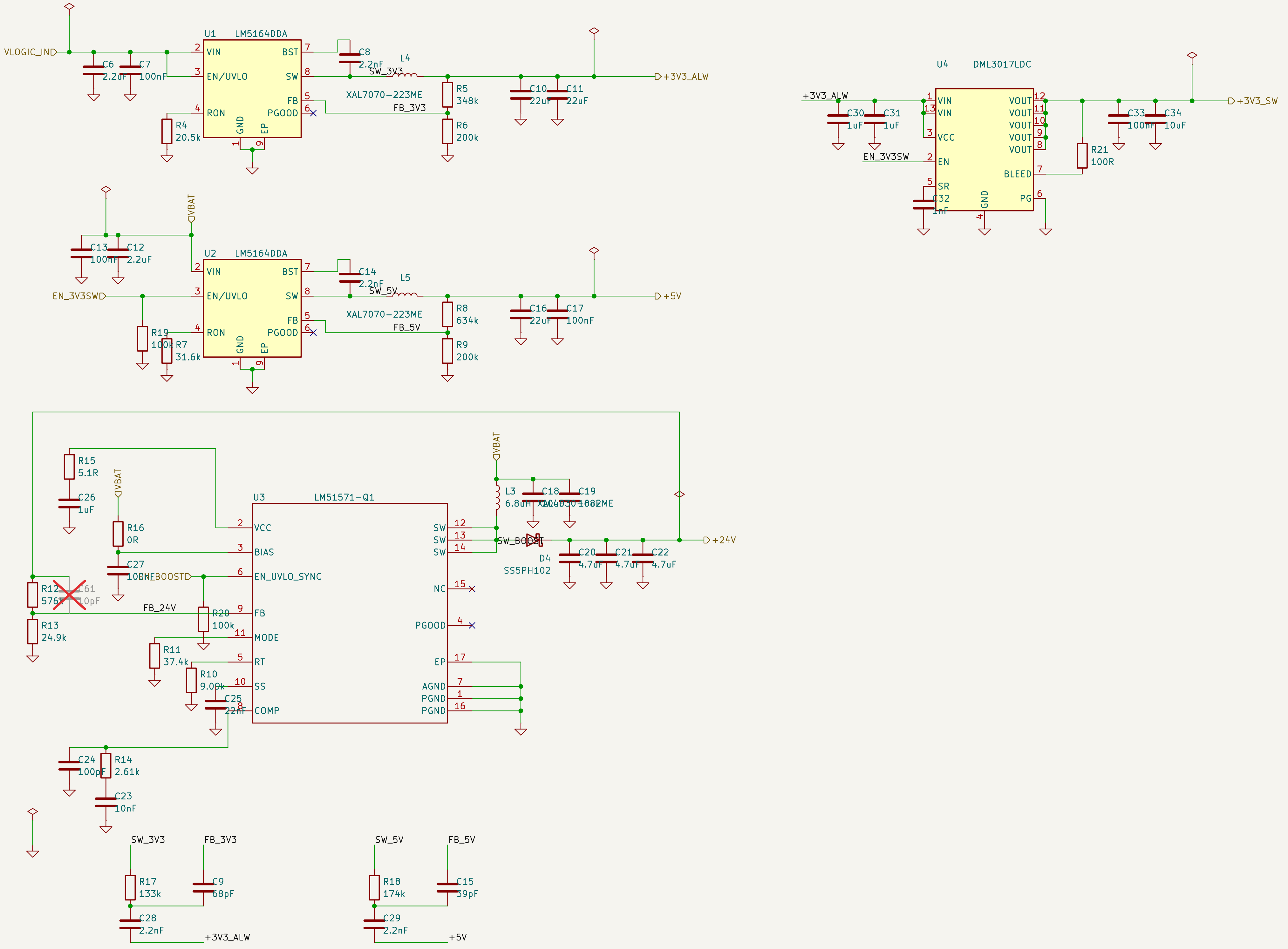
NET NAMED IGN_IN IS AN ADDITION TO THE CONTRACT

The plan says the divider is fed from "a VBAT-side input" but names no net. IGN_IN is the ignition-switched 12 V sense wire from the harness; it must land on a connector pin on the outputs sheet. Rename or remove it if the contract should stay closed.

EN_BOOST AND EN_3V3SW PULLDOWNS ARE ON THE RAILS SHEET, NOT HERE

The plan's Task 5 Step 4 asks for them here. They are already fitted as R19 and R20 next to the converters they gate, which is where they do the most good – the enable stays defined even if the inter-sheet net is open. One resistor per net; do not add a second.

Sheet: /mcu_can/
File: mcu_can.kicad_sch
Title: mcu_can
Size: A3 Date:
Rev:
KiCad E.D.A. 10.0.2
Id: 3/5



SLEEP-BUDGET CRITICAL VALUES -- DO NOT "TIDY"

Both feedback dividers are >= 500 kohm on purpose. The LM5164's quoted quiescent current does NOT include the external FB divider; a conventional 275 kohm divider would add ~12 uA per rail, unbudgeted. At 348k/200k and 634k/200k each draws 6.0 uA. A high-impedance FB node needs the feedforward cap for stability -- that is the trade being made deliberately.

EN_UVLO_SYNC has NO divider. A line-UVLO divider there would draw ~14 uA continuously whether the boost is enabled or not. Driving it straight from the GPIO keeps "floating = boost off" via the EN_BOOST pulldown.

U3 PIN 6 (EN_UVLO_SYNC) IS DRIVEN DIRECTLY BY EN_BOOST. THERE IS NO DIVIDER. DO NOT ADD ONE.

The LM51571-Q1 typical application puts RUVLOT/RUVLOB from VSUPPLY to this pin to set a line UVLO. That divider sits across the battery rail and conducts whenever the battery is present, enabled or not -- about 14 uA, a large fraction of this board's whole deep-sleep budget. It was removed on purpose: the MCU brown-out and the LM51571's own VCC UVLO (2.8 V typ, sourced from BIAS) already cover the low-battery case, so an external line UVLO buys nothing.

Consequences. (a) Pin 6 is high impedance with a ~5 uA hysteresis current source flowing OUT of it once above VUVLO. NOTHING on this sheet holds it low, so the EN_BOOST pulldown on mcu_can MUST exist -- 'floating = boost off' is only true if it is fitted. (b) ABSOLUTE MAXIMUM on pin 6 is 3.8 V to AGND (datasheet 8.1). A 3.3 V GPIO leaves 0.5 V of margin, so EN_BOOST must never see a 5 V pull-up. VUVLO(rising) is 1.5 V typ / 1.575 V max, so 3.3 V drives it hard enough. (c) Pin 6 is also the SYNC input; driving it DC means no external clock sync, which is intended.

2.2 MHz ON THE BOOST IS DELIBERATE: THE AM BROADCAST BAND IS 530 kHz - 1.7 MHz

This is a motorcycle. The boost is the loudest switcher on the board and it runs straight off the battery harness, which is the antenna. Its switching fundamental has to sit ABOVE the AM band, so U3 is set to 2.2 MHz -- the top of the LM51571-Q1's range and the frequency its electrical table is characterised at (RT = 9.09k; equation 5: RT = 2.21e10 / fRT - 955). R11 = 37.4k also enables Dual Random Spread Spectrum, whose low-frequency triangular profile targets exactly this band.

The two BUCKS CANNOT do the same thing: the LM5164's maximum switching frequency is 1 MHz, which is INSIDE the AM band. They are therefore placed BELOW it -- R4 = 20.5k gives 402 kHz on the 3.3 V rail, R7 = 31.6k gives 396 kHz on the 5 V rail (FSW[kHz] = VOUT[V] x 2500 / RRON[kohm]). Their fundamentals are clear of the band; harmonics 2 and 3 are not, and no LM5164 setting can fix that. Mitigate in layout and filtering, not by raising the frequency: keep both buck power loops tiny, and remember the common-mode choke on power_input is there for this noise.

THE LM5164 IS A CONSTANT-ON-TIME REGULATOR -- IT NEEDS IN-PHASE RIPPLE AT FB TO BE STABLE

COT has no error amplifier. Datasheet 6.3.1 / table 6-1 require at least 20 mV of ripple at FB that is IN PHASE with the inductor current. Ceramic output caps alone give capacitive ripple, 90 degrees out of phase; the classic failure is bursts of on-pulses followed by a long off-time. C9 / C15 are the type-2 feedforward caps the brief calls for -- they AC-couple output ripple to FB, but they do not by themselves make that ripple resistive. Finish it one of two ways before ordering boards.

(a) TYPE 2, as drawn: add a small series resistance in the output-cap leg. Equation 4 gives RESR >= 20 mV / dIL, i.e. ~67 mohm (3.3 V rail, dIL 0.30 A) and ~54 mohm (5 V rail, dIL 0.37 A) at the 22 uH / 400 kHz operating point. Only ripple current flows in it (<1 mW) so it is nearly free, but it must be a real part -- do not rely on the cap's own ESR. (b) TYPE 3: the RA / CA / CB injection network of figure 7-1 (453k / 3.3nF / 56pF in TI's 12 V design). That is what the EVM ships and it needs no series resistance at all.

C9 / C15 = 100 pF is well above the equation-6 minimum (3.1 pF, 2.7 pF) and puts the coupling zero near 4.5 kHz, two decades below FSW, so the full output ripple reaches FB.

WHERE EVERY VALUE CAME FROM

U1 / U2 LM5164DDA, VREF = 1.200 V, RFB2 = 1.2 / (VOUT - 1.2) x RFB1 (eq 10): 348k/200k -> 3.288 V, 634k/200k -> 5.004 V, 6.0 uA each. TI recommends RFB1 in 100k..1M, so both sit inside the datasheet's own range. RRON: R4 20.5k -> 402 kHz, R7 31.6k -> 396 kHz; minimum on-time at the worst case (VIN clamped to ~39 V by the input TVS) is 205 ns, well above the 50 ns floor. NOTE the LM5164 minimum VIN is 6 V, so a deep crank dip drops both bucks out -- correct, but know it. C8 / C14 = 2.2 nF is MANDATORY and exact: the datasheet says a larger bootstrap cap stresses the internal VCC regulator and damages the device. L4 / L5 = 22 uH gives 0.30 A (3.3 V) and 0.37 A (5 V) ripple at 13.8 V in, 30-37% of the 1 A rating; pick a shielded part whose saturation current clears the LM5164's 1.5 A peak current limit, not merely 1 A.

U3 LM51571-Q1, VREF = 1.000 V, VLOAD = VREF x (RFBT/RFBB + 1) (eq 9): 115k/4.99k -> 24.05 V (+0.2%), 200 uA, drawn only while the boost is enabled. RT 9.09k -> 2.2 MHz. MODE 37.4k -> hiccup overload protection ON and spread spectrum ON (9.4.3). SS 22 nF, COMP 2.61k + 10 nF, CHF 100 pF, VCC 5.1R + 1 uF, BIAS 0R + 100 nF are the datasheet's list-of-materials values for its 6-to-12 V / 1.6 A / 2.1 MHz boost example (table 10-2); the VCC 5.1R + 1 uF pair is a hard requirement of section 9.3.2, not a suggestion. THE COMPENSATION IS A STARTING POINT, NOT A RESULT -- re-run R14 / C23 / C24 through the LM5157x Boost Quick Start Calculator for 24 V and this board's real load before a board spin.

BIAS is fed from VBAT, not from +24V. Either is legal (BIAS operates 2.9 V to 45 V and the input TVS holds VBAT under that), but VBAT is unconditional whereas +24V only exists after the output pre-charges through L3 and D4. Shutdown current from BIAS is 2.6 uA typ / 5 uA max and would be drawn through L3 + D4 from VBAT anyway, so moving it saves nothing. Count that 2.6 uA in the sleep budget.

PGOOD IS LEFT OPEN ON ALL THREE REGULATORS -- NOT AN OVERSIGHT

PGOOD is open-drain and pulls LOW whenever its rail is down, which is precisely deep sleep. A 10k pull-up (the LM51571's recommended minimum) would sink 330 uA; even 100k sinks 33 uA continuously while asleep. No pull-up is fitted and no net is attached, so ERC reports these pins unconnected. No no-connect flags were added, on purpose: the ERC report is the reminder. If a rail-fault input is ever wanted, gate the pull-up -- do not just add one. U3 pin 15 (NC) is also left open.

GND ON THIS SHEET IS BOARD GROUND. There is no GND_IN here; this sheet is downstream of the common-mode choke on power_input, so every ground symbol is the board-side return. AGND (7), both PGND (1, 16) and the exposed pad (17) tie to that one net, so R10's 'resistor to AGND' lands on GND -- the same node. Star them properly in layout.

VBAT appears as THREE hierarchical labels (U2 input, L3 input, U3 BIAS feed) rather than one dragged wire. Same name, same sheet, same net -- confirmed in the exported netlist.

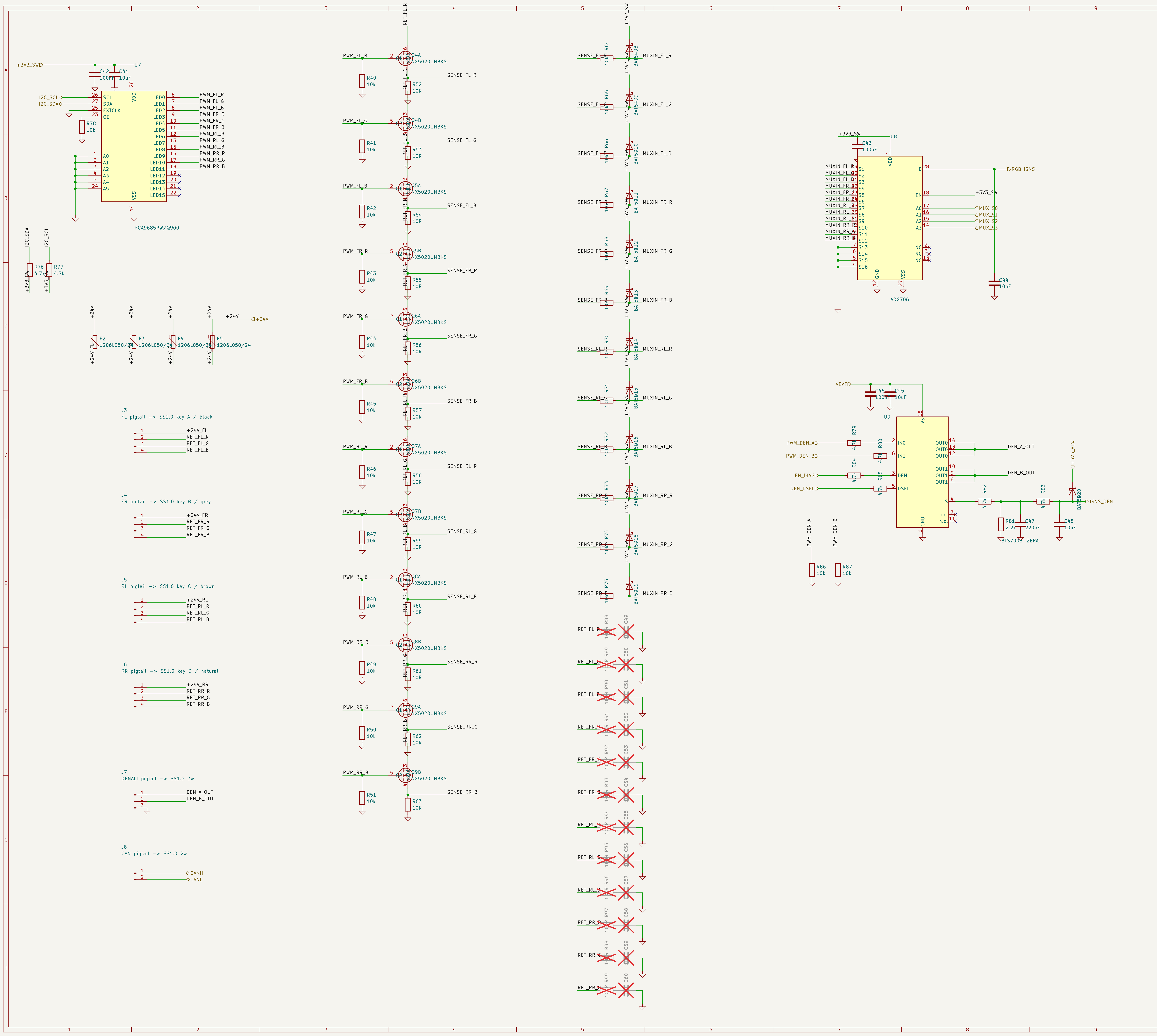
BEFORE THIS SHEET IS TURNED INTO A BOARD

1. No footprints are assigned to anything here; the Footprint field is empty on all 42 parts. 2. The 'rails' sheet symbol in mccan.kicad_sch has NO sheet pins, so all seven hierarchical labels below raise hier_label_mismatch in ERC -- add pins VLOGIC_IN (in), VBAT (in), EN_3V3SW (in), EN_BOOST (in), +3V3_ALW (out), +5V (out), +24V (out). 3. There is still no PWR_FLAG in the project, so GND reports power_pin_not_driven. 4. Decide type-2 vs type-3 ripple injection for the two bucks and fit the parts. 5. Re-run U3's compensation for 24 V. 6. Confirm the EN_BOOST pulldown exists on mcu_can and that nothing pulls EN_BOOST above 3.3 V. 7. L3 is 6.8 uH XAL4030-682ME -- check its saturation rating against the LM51571's 4.33 A switch current limit at the real 24 V load, not the nominal one. 8. D4 SS5PH102 sees VOUT plus switch-node overshoot; verify its rating against the real layout's ringing and keep the D4 / C20-C22 / PGND loop as small as physically possible -- at 2.2 MHz that loop is the whole EMI story.

Sheet: /rails/
File: rails.kicad_sch

Title: rails

Size: A3 Date:
Rev:
KiCad E.D.A. 10.0.2
Id: 4/5



ACCEPTANCE (spec 5, 9.1, 9.2)

- PCA9685 on +3V3_SW, I2C, LED0–11 -> the 12 RGB switch inputs IN ChannelIndex ORDER:
  LED0,1,2 = FL R,G,B; LED3,4,5 = FR R,G,B; LED6,7,8 = RL R,G,B; LED9,10,11 = RR R,G,B.
  LED12–15 UNUSED.
- 12x discrete logic-level N-MOSFET, gate direct from PCA9685, sense shunt in each source leg
- 12 SENSE_* nodes -> 16:1 analog mux (MUX_S0–S3) -> RGB_ISNS -> GPIO 32 (ADC1_CH4)
- NO SPI anywhere on this board
- Dual PROFET from VBAT, inputs PWM_DEN_A/B from ESP32 LEDC on GPIO 18/19, NOT the PCA9685
- PROFET: ONE sense output -> ISNS_DEN (GPIO 34) with scaling resistor; DEN_DSEL selects which channel IS reports
- 4x PTC, one per string, on the +24V feeds
- All 14 PWM_* nets have a pulldown to GND
- 4 corner connectors, keyed/colour-coded differently from each other
- Denali connector: 2 switched positives + 1 shared ground
- Output snubber footprints present but DNP

SENSE CHAIN (spec 5.1) –– VALUES REVISED, THE PLAN TEXT IS STALE

10 ohm x 26.7 mA = 267 mV at full channel current (133 mV if the 6 cm strip length is used). ADC1 at 0 dB attenuation (0–1.1 V range). Classification is open (~0 mV) / working (~267 mV) / shorted (saturated) –– NOT precision current metering.
Task 6 of the plan still says "1 ohm" and "140 mV", and asks for 2512 >= 1 W shunts. Those figures predate the RGB load being corrected downward by about 4x (spec 2.2); part-selection.md rows 1b.1–1b.3 carry the current values and explicitly WITHDRAW the 2512 requirement.
Sampling is on-demand: firmware drives one channel to 100%, steps the mux, reads, advances. The same sweep serves as the installation self-test.

SHUNT RATING DEPENDS ON A FIRMWARE BEHAVIOUR –– READ THIS BEFORE CHANGING EITHER

Normal dissipation in a 10 ohm shunt is 26.7 mA^2 x 10 = 7 mW, so an 0805 or 1206 is ample.
A SHORTED channel is different. The boost then current-limits at roughly 0.5 A, which puts about 2.5 W into that one shunt –– ten times a 1206's rating. It survives only because the diagnostic sweep is brief and because firmware LATCHES A SHORTED CHANNEL OFF once it classifies one. That makes the shunt package rating conditional on firmware behaviour, which is exactly the kind of cross-domain dependency that gets lost between disciplines.
It is written down in three places on purpose: here, in part-selection.md, and it belongs in the firmware spec (spec section 12) when that is written. If firmware does NOT latch off, move the shunts to 2512 1 W.

WHY ~OE IS PULLED LOW (R78) RATHER THAN HIGH

The plan says "OE tied so outputs are disabled by default". Taken literally that means pulling ~OE HIGH –– but no GPIO is allocated to release it, so the board would be permanently dark.
"Disabled by default" is already delivered twice over without this pin: the PCA9685 sits on +3V3_SW so it is unpowered in sleep, and it powers up with MODE1 SLEEP set and every LEDn register cleared, i.e. outputs off. R78 holds ~OE low so the part is usable; the safe state comes from the rail and the power-on register defaults.

SENSE-PATH PROTECTION (I15)

Each sense node reaches its mux input through 10 kOhm (R64–R75) with a clamp to +3V3_SW.
The SERIES RESISTOR is what actually bounds a fault: at 24 V on a sense node it admits 2 mA, far inside the ADG706's input rating. The clamps (D9–D20) are redundancy.
C44, 10 nF at the mux output, supplies the ADC's sample-and-hold charge locally, which is what makes 10 kOhm series resistors harmless. tau = 100 us, so firmware must wait >= 500 us after each mux step before reading. 100 nF would stretch that to 1 ms.
A quad diode array may replace D9–D20 at layout to save area.

PROFET U9 –– PINOUT VERIFIED, Infineon BTS7008-2EPA Rev. 1.21 Table 2 (2024-07-29)

EP/15 VS (exposed pad, the ONLY supply pin)   1 GND   2 IN0   3 DEN   4 IS   5 DSEL
6 IN1   7, 11 n.c. (internally not bonded)   8–10 OUT1   12–14 OUT0
Table 2 note 1: "All output pins of the channel must be connected together on the PCB." All three pins of each channel are therefore modelled and tied on this sheet, and the layout must carry the full channel current on every one of them.
FOOTPRINT (Task 8): Package_SO:Infineon_PG-TSDSO-14-22, which ships with KiCad. Its pad 15 is the 2.65 x 4 mm thermal pad, which is why VS is numbered 15 here. PG-TSDSO-14-22 is the former name of the same package –– the datasheet's revision history records the rename.

PROFET SENSE CHAIN –– HOW R81 WAS SIZED

kILIS = 5400 (Table 22, at IL14 = 2.8 A, +/-5.8%; 5450 at 5.5 A). At the Denali's 3.3 A per channel (spec 2.2) the IS pin therefore sources IL / kILIS = 611 uA.
R81 = 2.2 kOhm puts that at 1.34 V, comfortably mid-scale for ADC1 at 11 dB attenuation (useful range ~0.15 to 2.45 V on GPIO 34). Full scale then corresponds to 6.0 A –– above the 3.3 A working point, below the 7.5 A nominal rating, so a genuine overcurrent still reads on scale instead of pinning. The 0.15 V ADC floor corresponds to 368 mA, so an open channel (IL(OL) is ~21 mA, Table 22) reads as a hard zero and is unambiguous.
R82 (RIS_PROT) and R83 (RADC) are NOT optional. Under a fault the IS pin saturates: Table 20 gives IIS(SAT) = 4.1 to 15 mA and VSIS_F = VS – VIS of only 0.5 V typ / 1 V max, so the sense node sits within a volt of battery –– about 12.5 V. R83 is what keeps that off GPIO 34, with D21 clamping the remainder to +3V3_ALW at under 2 mA.
C47 (220 pF) with R83 gives the ~1 us time constant the datasheet asks for. C48 (10 nF) at the ADC end supplies the SAR sample-and-hold charge locally, which is what makes a 4.7 kOhm series resistor harmless; tau = 47 us, so firmware must settle >= 250 us after selecting a channel with DSEL before reading.

PROFET CONTROL LINES –– WHY THE PULLDOWNS ARE ON THE MCU SIDE

R79/R80/R84/R85 are the datasheet's RIN, RDEN and RDSEL (Table 24, 4.7 kOhm): they protect the MCU during overvoltage and reverse polarity, and they are what allows the outputs to switch OFF if the module loses its ground.
All four inputs must be held low while the ESP32 is in reset or deep sleep –– that is also what keeps the PROFET in Sleep mode (0.6 uA) rather than Stand-by, which needs INn, DEN and DSEL all low, and GPIO 18/19 float in reset. R86/R87 do that for the two PWM lines; EN_DIAG and DEN_DSEL already have R35 and R34 in mcu_can's passive-defaults block, so they are not duplicated here.
They are deliberately on the MCU side of the series resistors. In series the two would form a divider: 3.3 V x 10 / 14.7 = 2.24 V at the pin, against a guaranteed-high threshold of 2.0 V max (Table 7). A 240 mV margin over temperature and tolerance is not a margin.
On the MCU side the pin is driven through 4.7 kOhm into a microamp input, so the drop is negligible.

PROFET –– WHAT IS DELIBERATELY NOT FITTED

No RPD (47 kOhm output pulldown), ROL or T1. Those exist only for OFF-state open-load diagnosis, which spec 5.3 does not ask for; ON-state sense through IS covers what this design needs. RPD would also draw ~290 uA per channel whenever the output is on.

CORNER MIS-MATING –– STILL A SAFETY ITEM, BUT ONE MATING POINT FEWER (spec 9.2)

J3–J6 are NOT connectors. They are soldered 4-wire tails that leave the box through cable glands; the Superseal 1.0 4-way joints sit out on the harness. TE Superseal is a wire-to-wire series with no panel-mount board-side housing, so the panel connectors this sheet used to show could not have been built.
What that buys: the corner-to-board mapping is now fixed at ASSEMBLY and cannot be swapped at the box. Of the two places a front/rear swap could happen, the easy one –– four identical shells side by side on a panel –– is gone. Swapping at the far end is still possible, so the keying stays: FL key A / black, FR key B / grey, RL key C / brown, RR key D / natural, applied to the harness-side joint and the sleeve label. Spec 9.2's install self-test still applies.
Wire tails need the _Relief footprint variant and they need it fitted properly: the second, unplated hole per conductor is what takes a pull instead of the pad. Nothing else mechanically restrains these wires.

PTCs (F2–F5), one per string

Isolates a shorted string so one crushed cable cannot extinguish all four corners (spec 5.2).
Littelfuse 1206L050/24: Ihold 0.50 A at 23 C, 0.31 A at 65 C, against an 80 mA per-string load.
That is a 3.9x margin at the hot internal temperature, which is the case that matters.

Sheet: /outputs/
File: outputs.kicad_sch
Title: outputs
Size: A2
Date:
Rev:
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